

Session 5 - Regularisation and model selection

Foundations of Machine Learning (Demo), E1234

6 October 2026

The question

You can fit a straight line, a cubic, or a degree-15 polynomial. All three minimise training error to different degrees, and the most flexible always wins on the training set. So how do you choose? Not by fitting better - by estimating out-of-sample error honestly.

Bias-variance, made concrete

Expected squared error at a point decomposes as:

$$E[(y - \hat{f}(x))^2] = \text{bias}(\hat{f})^2 + \text{var}(\hat{f}) + \sigma^2$$

- **Bias:** error from a family too rigid to contain the truth. Falls as flexibility rises.
- **Variance:** how much $\hat{f}$ would change with a different sample. Rises as flexibility rises.
- **σ^2 :** irreducible noise. No model beats this, and claiming to means leakage.

The sum is U-shaped in flexibility. Model selection is finding the bottom of that U without peeking at the test set.

Regularisation: pay for complexity

Add a penalty on the size of the coefficients:

Ridge (L2): $\min ||y - Xb||^2 + \lambda \sum_j b_j^2$

Lasso (L1): $\min ||y - Xb||^2 + \lambda \sum_j |b_j|$

- $\lambda = 0$: ordinary least squares. $\lambda \rightarrow \infty$: all coefficients to zero.
- **Ridge** shrinks coefficients smoothly, keeps all features, and rescues collinear designs
 - $X'X + \lambda I$ is invertible even when $X'X$ is not.
- **Lasso** sets some coefficients exactly to zero: it selects. The corner of the L1 ball is what makes exact zeros possible.
- **Elastic net** interpolates, and is the safe default with correlated features.

Never penalise the intercept, and always standardise first - a penalty on raw coefficients otherwise means “penalise whichever feature happens to be measured in small units”.

Validation

- **Hold-out:** one split. Fast, and noisy on small n .
- **k-fold cross-validation:** split into k folds, train on $k-1$, validate on the remaining one, rotate, average. $k = 5$ or 10 . The standard tool.
- **Stratified k-fold:** preserve class proportions - mandatory for imbalanced classification.
- **Grouped / time-series splits:** when rows are not exchangeable. Repeated measurements on the same municipality must not straddle a fold; future must not predict past.

The selection procedure

1. Set the test set aside. Do not look at it.

2. For each candidate λ , run k-fold CV on the training set; average the fold scores.
3. Pick the λ with the best mean CV score - or, better, the **one-standard-error rule**: the simplest model within one standard error of the best. It costs almost nothing in accuracy and buys real stability.
4. Refit on the whole training set at that λ .
5. Report the test score once.

Nested CV, and why people get this wrong

If you use CV both to tune and to report, the reported number is optimistic - the folds have seen the tuning decisions. When you need an unbiased performance estimate *and* tuning, use nested CV: an inner loop to select, an outer loop to evaluate. Expensive, and correct.

Learning curves read the diagnosis

Plot train and validation error against training-set size:

- both high, converged together -> **underfitting**. More data will not help; a richer model or better features will.
- large persistent gap -> **overfitting**. More data or more regularisation will help.

Two lines on one plot answers the question “should I collect more data?” - which is usually a budget question.

Information criteria, in passing

AIC and BIC penalise fitted likelihood by parameter count and require no resampling. Useful for nested parametric models, awkward for the rest. Cross-validation is more general and assumes less; prefer it unless n is tiny.

Today’s demo and lab

- `demo.py`: polynomial degrees 1, 3 and 15 fitted to 20 noisy points, printing train and validation RMSE side by side - the U-shape in three lines of output. Then a ridge path over the degree-15 family alone, which recovers a better validation score than any unpenalised degree.
- `labs/05_regularisation-and-model-selection`: RidgeCV and LassoCV over a λ grid, the CV curve plotted, and the lasso’s zeroed coefficients read off as a feature selection you have to justify.

Assignment 2 is released today

Classification and evaluation, as a notebook: implement the sigmoid, log-loss and the confusion-matrix metrics, then choose and defend a threshold. Due 3 November, 23:59.